



Udharwale

A Personal Ledger & Debt Tracking Web App

Capstone Project Report · MCA · Lovely Professional University Online

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The Problem: Informal Debt Goes Untracked

In many communities, borrowing small amounts, splitting bills, or buying on credit from local shops is everyday reality. Yet these micro-transactions are tracked on paper or memory – leading to **errors, confusion, and strained relationships**.

Individuals

Track money lent to or borrowed from friends and family.

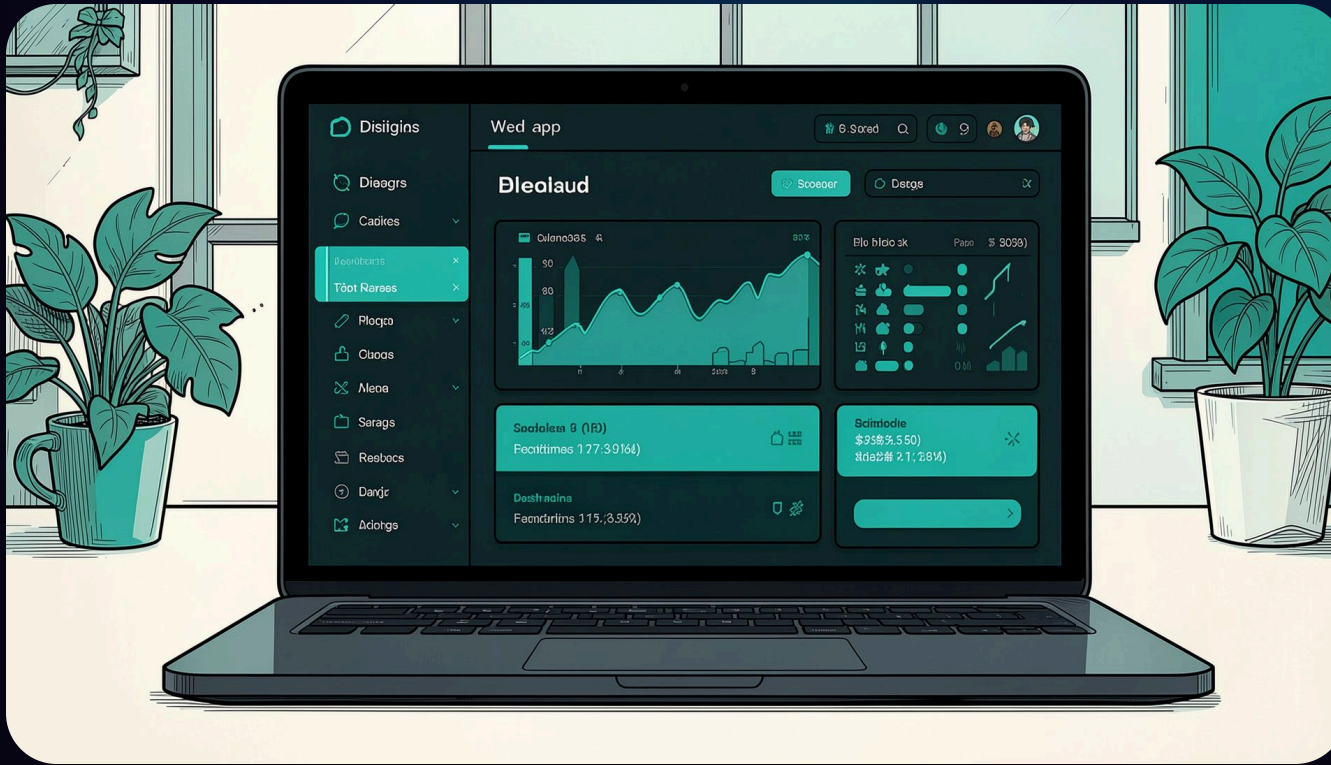
Shopkeepers

Keep a digital "Khata" of customers who buy on credit.

Students

Manage shared expenses like rent, food, and travel.

What is Udharwale?



Udharwale digitizes personal debt management — replacing paper ledgers with a secure, intuitive web application.

→ Categorize Transactions

Log as "Gave" (Lent) or "Got" (Borrowed).

→ Dynamic Dashboard

Real-time overview of total balances and net cashflow.

→ Secure Authentication

Login with email, password, and PIN recovery.

Existing Solutions & The Gap



Why Udharwale?

Splitwise is too group-focused. **KhataBook** is too business-oriented. Neither serves the everyday individual who simply wants to track "I gave X" and "I got Y" transactions.

- Udharwale fills this gap with a minimalist, one-on-one interface accessible to users with minimal technical literacy.

Tools & Technologies



Frontend

Next.js 16 (App Router), React 19, Tailwind CSS 4



Backend

Next.js API Routes in a Node.js environment



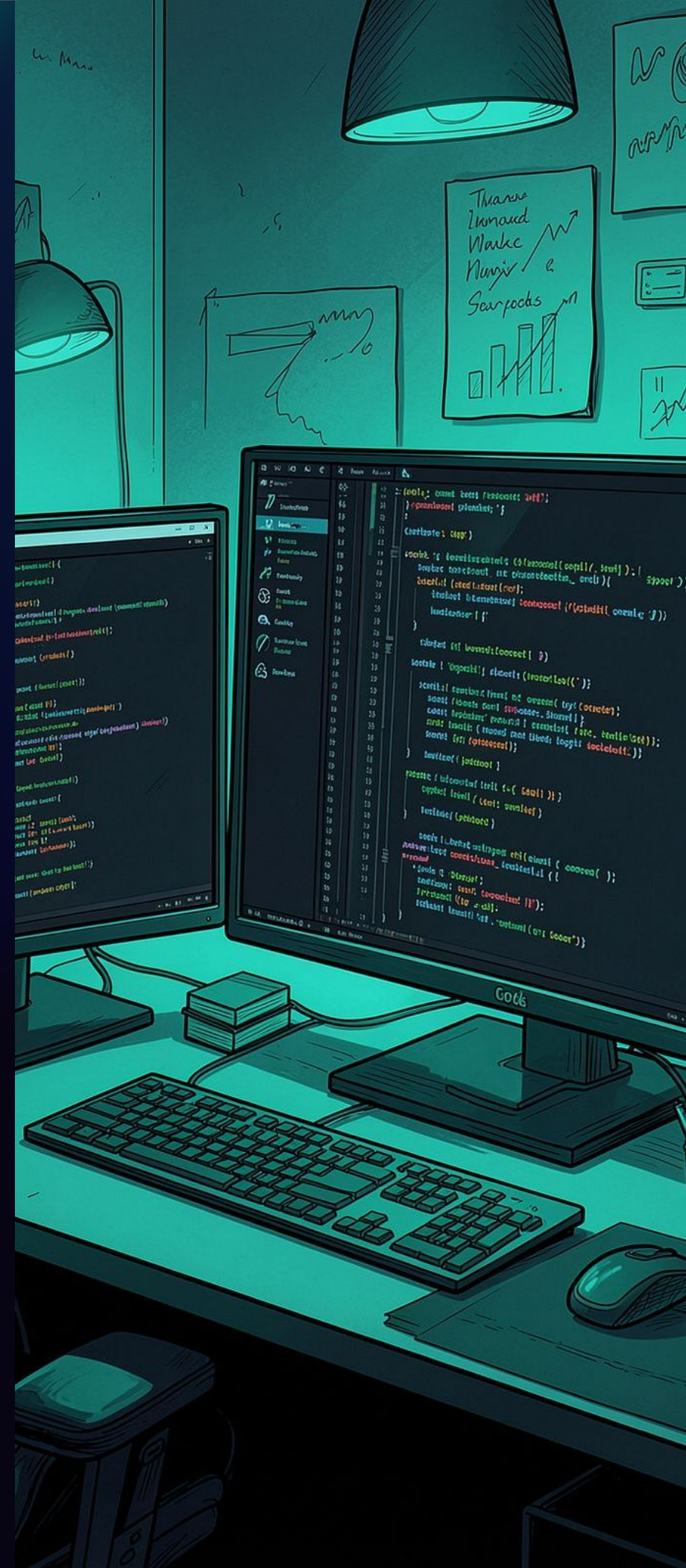
Database

MongoDB via Mongoose – flexible document schemas



Language

TypeScript for strong typing and error reduction



Database Design

Two Primary Collections

Users

Stores name, email, hashed password, **recovery PIN**, and security answer.

Contacts

Mapped to userId. Stores contact details (name, phone, email) with an **embedded array of transactions** — each holding amount, type ('gave'/'got'), description, date, and category.



System Modules



Authentication Module

Handles secure login, registration, and password recovery using a custom PIN and security question.



Dashboard Module

Fetches and aggregates all transactions to display total outstanding balances in real time.



Contact Management

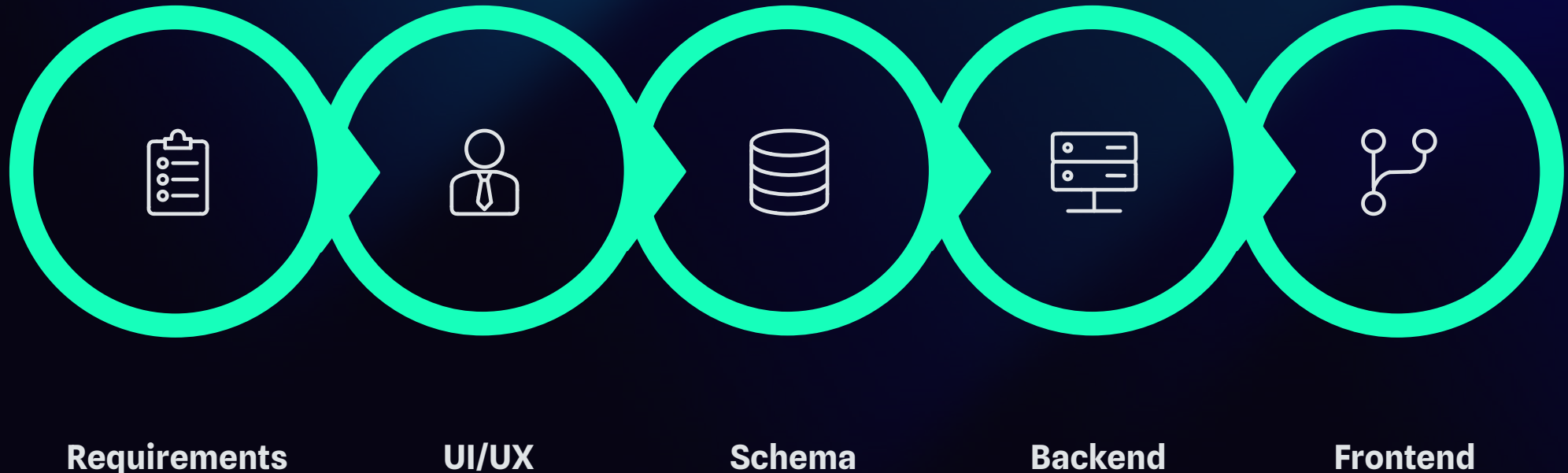
Allows users to add, edit, and view individual contacts with full transaction history.



Transaction Module

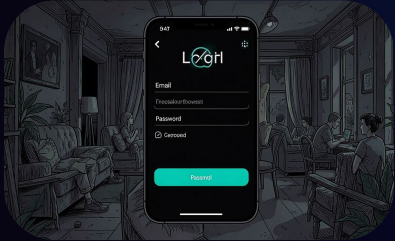
Enables adding new financial records tied to specific contacts and updates aggregated totals instantly.

Methodology: Agile Development



Iterative development allowed continuous testing and rapid incorporation of UI/UX improvements throughout the project lifecycle.

Results: The Application in Action



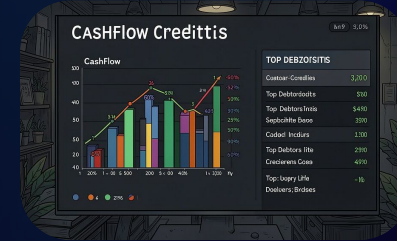
Login & Registration

Clean, welcoming authentication screens with PIN-based account recovery backup.



Main Dashboard

Accurately calculates net balance.
Green for money to receive, **Red** for money owed.



Overview View

Displays total to receive, total owed, net cashflow ratio, and distribution by category.

Embedding transactions within the Contact model significantly reduced database query times. The app is fully responsive on mobile browsers.

Conclusion & Future Scope

Conclusion

Udharwale successfully demonstrates how modern web technologies can solve a practical, everyday problem — reducing the cognitive load of remembering informal debts and minimizing financial disputes among peers.

Future Enhancements

01

WhatsApp/SMS Integration

Automated reminders for pending payments.

02

Multi-Currency Support

Log transactions in different currencies.

03

Data Export

Export histories to PDF or Excel.

04

Progressive Web App (PWA)

Install natively on smartphones.